```
// Name : Tanushree Lohiya
// Roll no. : SEA56
// Graph Traversal for Campus Navigation Using BFS and  DFS.
#include <stdio.h>
int g[10][10], visited[10], n;
void DFS(int v)
{
 int i;
 printf("%d ", v);
 visited[v] = 1;
 for(i = 0; i < n; i++)
 if(g[v][i] && !visited[i])
 DFS(i);
}
void BFS(int start)
{
 int q[10], front = 0, rear = 0, i, v;
 for(i = 0; i < n; i++)
      visited[i] = 0;
 q[rear++] = start;
 visited[start] = 1;
 while(front < rear)
 {
```

```
 v = q[front++];
printf("%d ", v);
 for(i = 0; i < n; i++)
 {
 if(g[v][i] && !visited[i])
 {
 q[rear++] = i;
 visited[i] = 1;
 }
 }
 }
}
int main()
{
 int e, i, j, u, v, start;
 printf("Enter number of Buildings (max 10): ");  scanf("%d", &n);
 if(n < 1 || n > 10)
 return 0;
 printf("Enter number of Paths: ");
 scanf("%d", &e);
 printf("Enter Actual path:\n");
 for(i = 0; i < e; i++)
```

```
 {
 scanf("%d%d", &u, &v);
g[u][v] = g[v][u] = 1;
 }

printf("\nAdjacency Matrix:\n");
 for(i = 0; i < n; i++)
 {
 for(j = 0; j < n; j++)
 printf("%d ", g[i][j]);
 printf("\n");
 }
 printf("Enter starting Buildings: ");
 scanf("%d", &start);
 for(i = 0; i < n; i++)
      visited[i] = 0;
 printf("DFS: ");
 DFS(start);
 printf("\nBFS: ");
 BFS(start);
 return 0;
}
```

Output :

```
Enter number of Buildings (max 10): 5
```

Enter number of Paths: 6

Enter Actual path:

0 1

0 2

0 3

1 3

3 4

2 4

Adjacency Matrix:

0 1 1 1 0

1 0 0 1 0

1 0 0 0 1

1 1 0 0 1

0 0 1 1 0

Enter starting Buildings: 0

DFS: 0 1 3 4 2

BFS: 0 1 2 3 4

Process returned 0 (0x0)   execution time : 32.475 s Press any key to continue.